

University Travel Data Analysis

Illustrative flights, rail, car, taxi and hotel activity transformed into travel-emissions and management KPIs.

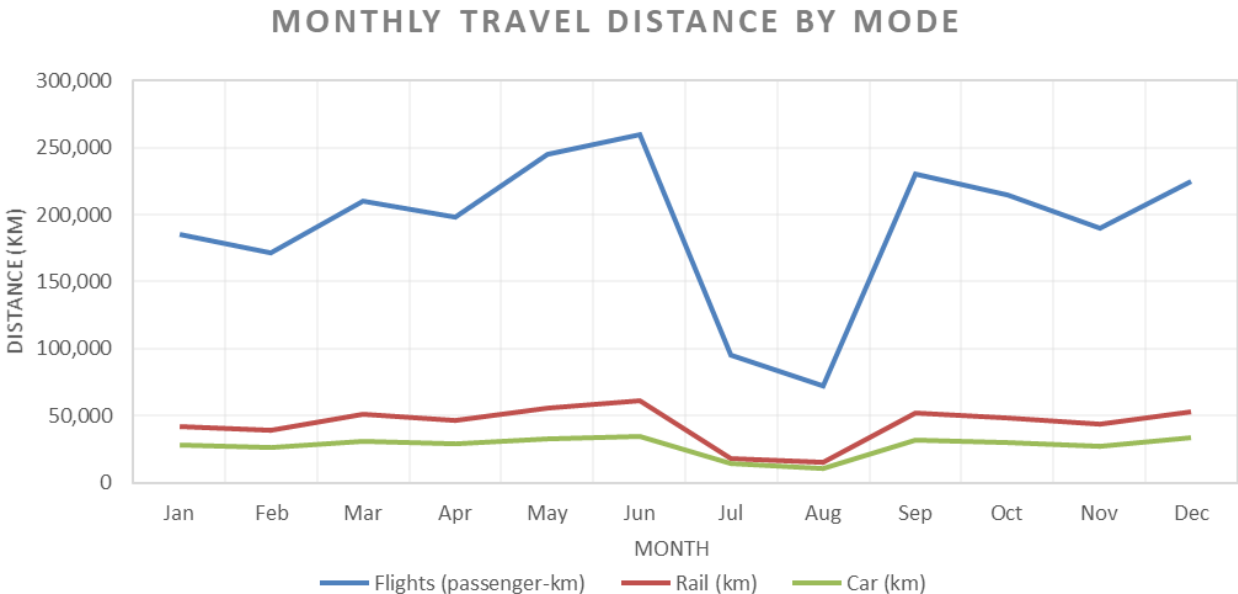
Executive summary

This synthetic case demonstrates the workflow from raw operational data to a management-ready analysis. All figures are illustrative and not real client data.

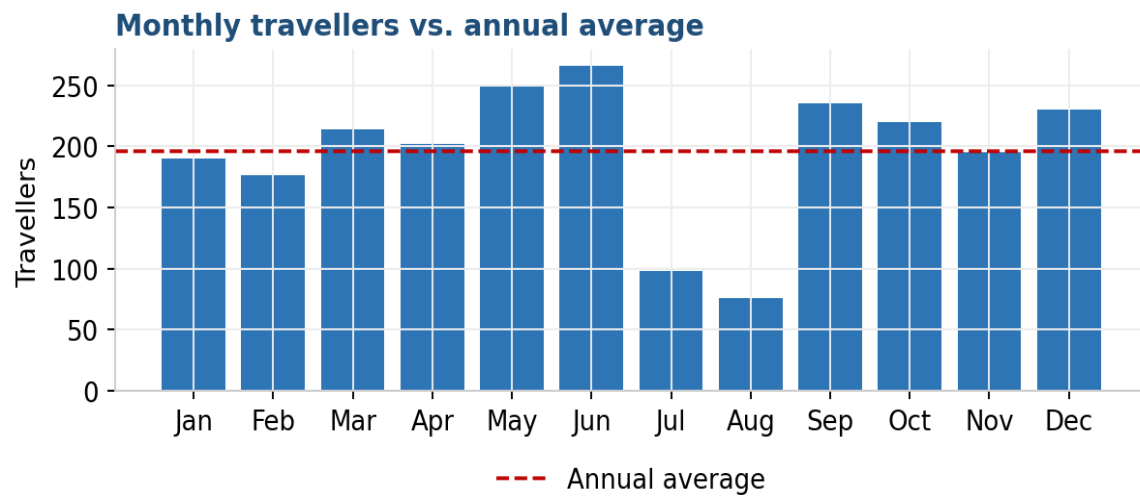
KPI	Illustrative result	Unit
Travel emissions	551.6	tCO2e
Flight distance	2.30	million passenger-km
Air share of emissions	75.0	%
Emissions / traveller	234.5	kgCO2e

Activity analysis

Air travel is the dominant emissions category in this illustrative dataset. Combining distance, mode, hotel nights and traveller count gives far more management context than a single annual total.



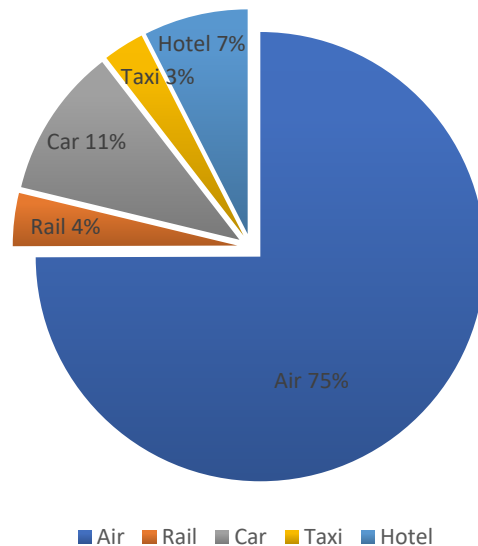
Travel volume tracks the academic calendar closely: traveller counts fall 55.6% below the annual average in July/August, with June the peak month.



Carbon / KPI analysis

Illustrative factors are simple screening assumptions and no aviation radiative-forcing uplift is applied. A robust implementation would reconcile booking, expense and reimbursement sources.

Annual travel emissions by mode (tCO2e)



Key insight: air travel accounts for 75% of annual travel emissions despite being only one of five modes tracked. This insight could turn a single "tonnes of CO2e" figure into something a travel policy can actually target and minimise.

Data quality

This is to determine whether staff, students and visitors are included; document exclusions and data gaps; align with the organisation's agreed methodology. As a built-in consistency check, hotel nights run 0.98 per traveller in this dataset, close to the expected ~1:1 ratio for single-night-equivalent trips.

Recommendations

1. Reconcile booking and expense data sources to confirm traveller counts and trip lengths before reporting externally.
2. From the results, air travel policy (booking class, route consolidation, virtual-meeting defaults) is where the emissions reduction opportunity actually lie.
3. Further analysis such as segmenting travel by purpose or organisational unit (research, admin, student exchange) can be used to target any reduction policy where it will have the most effect.
4. Hotel-nights-per-traveller ratio as an ongoing data-quality check each reporting period.

Scope note: This portfolio demonstrates data preparation, screening calculations, KPI development and reporting. It is not independent GHG verification or assurance, a statutory energy audit, legal advice, or a guarantee of savings. All figures in this document are illustrative, synthetic data for demonstration purposes only.